



Birla Institute of Technology & Science, Pilani

Work-Integrated Learning Programmes Division
MTech in Artificial Intelligence and Machine Learning
S2_2025-2026, AIMLCZG557/AECLZG557

Assignment 1 – PS2 - Submarine - [Weightage 12%]

Read through this entire document very carefully before you start!

Problem statement

Assume a submarine agent deployed at a specified location on an oceanic grid. A distress signal has been detected from a sunken vessel at another location on the ocean floor. Help the submarine agent navigate to the distress location to initiate a rescue operation.

The problem involves guiding the submarine through an underwater grid that has varying sea conditions and hazards: coral reefs, deep trenches, underwater current corridors, sonar blackout zones, and thermal vents.

The following constraints govern the submarine's movement through the grid:

- Thermal vent regions are completely impassable — extreme heat causes immediate hull damage. These cells act as hard walls and the submarine cannot enter them under any circumstance.
- Sonar blackout zones are to be avoided unless absolutely no alternative path exists — navigating without sonar significantly increases the risk of collision with underwater structures.
- Coral reef patches cost double the normal movement — the submarine must reduce to minimum speed, consuming twice the fuel and time to cross.
- Underwater current corridors cost half the normal movement — the natural flow assists propulsion, cutting fuel consumption and travel time in half.

- Deep trench zones are traversable but require the submarine to descend an additional depth level, adding a fixed overhead of 2 extra movement units to any trench cell.

The grid below gives the initial position of the submarine (start) and the sunken vessel (goal), along with the placement of all ocean terrain types across the grid. The environment is fully observable.

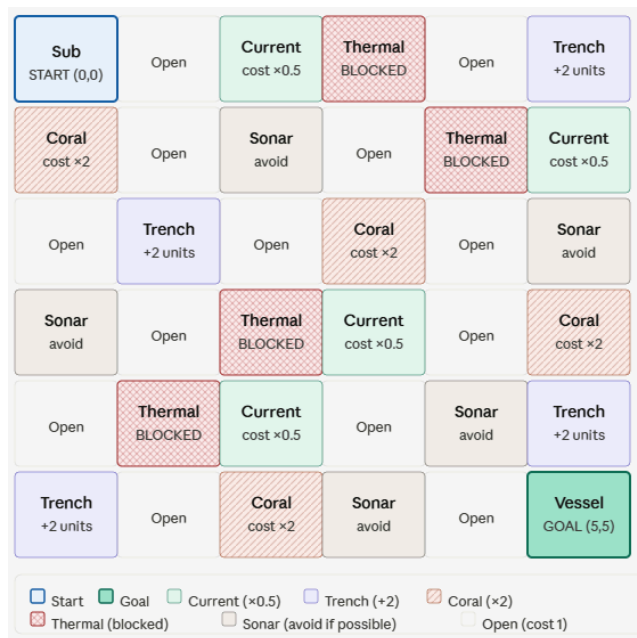
The desired solution must reach the goal through the least costly path, respecting all movement constraints.

The algorithm must accept the destination from the user and determine the most efficient route using intelligent search techniques.

Case 1:

Grid Size: 6 x 6

Grid Representation:



Legend:

S = Start Position

G = Goal Position

O = Open Water (cost = 1)

R = Coral Reef (cost = 2)

C = Underwater Current Corridor (cost = 0.5)

D = Deep Trench (cost = 3)

B = Sonar Blackout Zone (avoid if possible)

T = Thermal Vent (Blocked)

Sample input:

Source: (0,0)

Destination: (5,5)

Grid Size: 5 x 5

Sample output:

Best Path:

(0,0) → (0,1) → (1,1) → (1,2) → (1,3) → (2,4) → (3,4) → (4,3) → (5,4) → (5,5)

Total Grid Cells Traversed: 10

Total Movement Cost: 10.5

Case 2:

Grid Representation:



Legend:

S = Start Position

G = Goal Position

O = Open Water (cost = 1)
R = Coral Reef (cost = 2)
C = Underwater Current Corridor (cost = 0.5)
D = Deep Trench (cost = 3)
B = Sonar Blackout Zone (avoid if possible)
T = Thermal Vent (Blocked)

Sample input:

Source: (0,0)

Destination: (4,4)

Sample output:

Best Path:

(0,0) → (0,1) → (1,2) → (2,2) → (2,3) → (3,4) → (4,4)

Total Grid Cells Traversed: 7

Total Movement Cost: 5.5

- ☐ Explain the heuristic that can be used to solve this problem. Justify your choice with respect to the ocean grid and movement constraints.
- ☐ Explain the cost function associated with each terrain type and how it influences the A* search in reaching the goal.
- ☐ Explain why informed search algorithm A* is suitable for this constraint-based underwater navigation problem.
- ☐ Implement the A* algorithm in Python, representing the ocean grid, terrain constraints, and movement costs accurately.
- ☐ Print the following outputs from your implementation:
 - The complete path taken by the submarine from start (0,0) to goal (5,5), listing each grid cell visited in order.
 - The terrain type encountered at each cell along the path.
 - The movement cost incurred at each step.
 - The total number of grid cells traversed.
 - The total accumulated movement cost of the final path.

The assignment will be evaluated on the following points

1. PEAS analysis
2. Explanation of the heuristic and algorithm chosen
3. Implementation of the informed search in PYTHON
4. Print the details generated by the algorithm.

Deliverables:

- PDF document **designPSXX_<group id>.pdf** detailing your solution.
- **[Group id] _Contribution.xlsx** mentioning the contribution of each student in terms of percentage of work done. Columns must be “Student Registration Number”, “Name”, “Percentage of contribution out of 100%”. If a student did not contribute at all, it will be 0%, if all contributed then 100% for all.
- **inputPSXX.txt** file used for testing
- **outputPSXX.txt** file generated while testing
- .py file containing the python code. Create a single notebook. Do not fragment your code into multiple files
- Zip all of the above files including the design document in a folder with the name:
 - **[Group id] _A1_PSXX_XXXXXXXXXX.zip** and submit the zipped file.
 - Group Id should be given as **Gxxx** where **xxx** is your group number. For example, if your group is 26, then you will enter **G026** as your group id.

Instructions:

- It is compulsory to make use of the data structure(s) / algorithms mentioned in the problem statement.
- Ensure that all data structure insert and delete operations throw appropriate messages when their capacity is empty or full. Also ensure basic error handling is implemented.
- For the purposes of testing, you may implement some functions to print the data structures or other test data. But all such functions must be commented before submission.
- Make sure that you read, understand, and follow all the instructions
- Ensure that the input, prompt and output file guidelines are adhered to. Deviations from the mentioned formats will not be entertained.
- The input, prompt and output samples shown here are only a representation of the syntax to be used. Actual files used to evaluate the submissions will be different. Hence, do not

hard code any values into the code.

- Please note that the design document must include:
 - One alternate way of modeling the problem with the performance implications.
 - Writing a good technical report and well documented code is an art. Your report cannot exceed 4 pages. Your code must be modular and quite well documented.
 - You may ask queries in [this dedicated discussion section](#). Beware that only hints will be provided and queries asked **in other channels like MS Teams, emails, Taxila inbox will not be responded to.**

Deadline

1. The strict deadline for submission of the assignment is **8th June 2026 11:55 PM IST (Monday)**.
2. The deadline has been set considering extra days from the regular duration in order to accommodate any challenges you might face. No further extensions will be entertained.
3. Late submissions will be evaluated as below. Submissions made between:
 - a. 8th June 2026 11:56PM to 9th June 2026 11:55PM → Deduct 2M for late submission and evaluation for 10 marks.
 - b. 9th June 2026 11:56PM to 10th June 2026 11:55PM → Deduct 6M for late submission and evaluation for 6marks.
 - c. Beyond 10th June 2026 11:55PM → No evaluations, 0 marks will be awarded.

How to submit

- This is a group assignment.
- Each group has to make one submission (only one, no resubmission) of solutions.
- Each group should zip all the deliverables in one zip file and name the zipped file as [Group id] _A1_PSXX_XXXXXXXXXX.zip and submit the zipped file.
- Group Id should be given as Gxxx where xxx is your group number. For example, if your group is 26, then you will enter G026 as your group id.
- Assignments should be submitted via Taxila > Assignment section. Assignments submitted via other means like email etc. will not be graded.

Evaluation

- The assignment carries **12 Marks**.
- Grading will depend on:

- Fully executable code with all functionality working as expected
 - Well-structured and commented code
 - Accuracy of the design document.
 - Every bug in the functionality will have negative marking.
 - Marks will be deducted if your program fails to read the input file used for evaluation due to change / deviation from the required syntax.
- We encourage students to take the upcoming assignments and examinations seriously and submit only original work. Please note that plagiarism in assignments will be taken seriously. Refer to the detailed policy [here](#).
- Source code files which contain compilation errors will get at most 25% of the value of that question.

Readings

Artificial Intelligence: A Modern Approach Textbook by Peter Norvig and Stuart J. Russell, 4th Edition.